

Elligator

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Chapter 1

Elligator 1

Theorem 1.

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Theorem 2.

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Theorem 3.

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Theorem 4.

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Theorem 127 (u defined). *The following element of $\mathbb{F}_q$ is defined for each $t \in \mathbb{F}_q \setminus \{-1, 1\}$:*

$$u = (1 - t)/(1 + t),$$

Proof. □

Theorem 128 (Y defined). *The following element of $\mathbb{F}_q$ is defined for each $t \in \mathbb{F}_q \setminus \{-1, 1\}$:*

$$Y = (\chi(v)v)^{(q+1)/4} \chi(v) \chi(u^2 + 1/c^2),$$

Proof. □

Theorem 129 (x defined). *The following element of $\mathbb{F}_q$ is defined for each $t \in \mathbb{F}_q \setminus \{-1, 1\}$:*

$$x = (c - 1)sX(1 + X)/Y,$$

Proof. □

Theorem 130 (y defined). *The following element of $\mathbb{F}_q$ is defined for each $t \in \mathbb{F}_q \setminus \{-1, 1\}$:*

$$y = (rX - (1 + X)^2)/(rX + (1 + X)^2).$$

Proof. □

Theorem 131 (Fulfill Helper Curve Equation). *Let r , X and Y as defined above.*

Then $Y^2 = X^5 + (r^2 - 2)X^3 + X$ holds.

Proof. □

Theorem 132 (Non zero helper variables). *Let u , v , X , Y , x , y as defined above.*

Then $uvXYx(y + 1) \neq 0$.

Proof. □

Theorem 133 (Fulfill Curve Equation). *Let x and y as defined above.*

Then $x^2 + y^2 = 1 + dx^2y^2$ holds.

Proof. □

Definition 134.

Definition 135 (The Map). *Definition 2. In the situation of Theorem 1, the decoding function for the complete Edwards curve $E : x^2 + y^2 = 1 + dx^2y^2$ is the function $\varphi : \mathbb{F}_q \rightarrow E(\mathbb{F}_q)$ defined as follows:*

$$\varphi(\pm 1) = (0, 1);$$

if $t \notin \{\pm 1\}$ then

$$\varphi(t) = (x, y).$$

Original: Chapter "3.2 The map": Definition 2

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Theorem 226.	

Proof.

□

Definition 227 (The Encoding Function). In the situation of Definition 2, assume that q is prime, and define $b = \lfloor \log_2 q \rfloor$. Define $\sigma : \{0, 1\}^b \rightarrow \mathbb{F}_q$ by

$$\sigma(\tau_0, \tau_1, \dots, \tau_{b-1}) = \sum_i \tau_i 2^i.$$

Define

$$S = \sigma^{-1}(\{0, 1, 2, \dots, (q-1)/2\}).$$

Define $\iota : S \rightarrow E(\mathbb{F}_q)$ as follows:

$$\iota(\tau) = \varphi(\sigma(\tau)).$$

Original: Chapter "3.4 Encoding as strings": Theorem 4

Theorem 228 (Cardinality of S). In the situation of Definition 2, assume that q is prime, and define $b = \lfloor \log_2 q \rfloor$. Define $\sigma : \{0, 1\}^b \rightarrow \mathbb{F}_q$ by

$$\sigma(\tau_0, \tau_1, \dots, \tau_{b-1}) = \sum_i \tau_i 2^i.$$

Define

$$S = \sigma^{-1}(\{0, 1, 2, \dots, (q-1)/2\}).$$

Then $\#S = (q+1)/2$;

Original: Chapter "3.4 Encoding as strings": 1. statement of Theorem 4

Proof.

□

Theorem 229 (Cardinality of S). In the situation of Definition 2, assume that q is prime, and define $b = \lfloor \log_2 q \rfloor$. Define $\sigma : \{0, 1\}^b \rightarrow \mathbb{F}_q$ by

$$\sigma(\tau_0, \tau_1, \dots, \tau_{b-1}) = \sum_i \tau_i 2^i.$$

Define

$$S = \sigma^{-1}(\{0, 1, 2, \dots, (q-1)/2\}).$$

Define $\iota : S \rightarrow E(\mathbb{F}_q)$ as follows:

$$\iota(\tau) = \varphi(\sigma(\tau)).$$

Then $\#S = (q+1)/2$;

Original: Chapter "3.4 Encoding as strings": 1. statement of Theorem 4

Proof.

□

Theorem 230.

Proof.

□